



AXI Power Consumption Information

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Voltage [V]				Logging Information	
	AN	BN	CN		
Max	236.1 V 11/4/2015 2:55:04 PM	234.6 V 11/4/2015 2:55:12 PM	235.5 V 11/4/2015 2:55:03 PM	Study type: Energy study Topology: 3-ph Wye Start date: 11/4/2015 2:10:37 PM End date: 11/4/2015 3:10:37 PM Duration: 1h 0m 0s Averaging interval: 1sec Number of averaging intervals: 3600 (3600) * ... series contained invalid values that have been discarded for the shown result.	
Avg	234.7 V	233.1 V	234.0 V		
Min	230.5 V 11/4/2015 2:17:51 PM	229.0 V 11/4/2015 2:17:51 PM	229.9 V 11/4/2015 2:17:51 PM		
Current [A]					
	A	B	C		
Max	1.635 A 11/4/2015 3:03:31 PM	3.564 A 11/4/2015 2:14:41 PM	3.988 A 11/4/2015 2:42:16 PM		
Avg	1.090 A	2.062 A	0.783 A		
Min	0.456 A 11/4/2015 2:34:12 PM	1.458 A 11/4/2015 2:59:22 PM	0.367 A 11/4/2015 2:18:25 PM		
Frequency [Hz]					
	AN	BN	CN		
Max	50.03 Hz 11/4/2015 2:55:20 PM				
Avg	49.94 Hz				
Min	49.89 Hz 11/4/2015 2:20:06 PM				
THD-V [%]					
	AN	BN	CN		
Max	2.0 % 11/4/2015 2:11:42 PM	1.9 % 11/4/2015 2:58:35 PM	2.1 % 11/4/2015 2:11:42 PM		
Avg	1.5 %	1.3 %	1.5 %		
Min	1.3 % 11/4/2015 2:54:03 PM	1.2 % 11/4/2015 2:44:29 PM	1.3 % 11/4/2015 2:44:15 PM		
THD-A [%]					
	A	B	C		
Max	60.6* % 11/4/2015 2:39:25 PM	47.2 % 11/4/2015 3:09:46 PM	85.8* % 11/4/2015 2:34:47 PM		
Avg	59.2** %	22.1 %	202.7** %		
Min	45.4* % 11/4/2015 2:22:01 PM	19.2 % 11/4/2015 2:14:24 PM	76.0* % 11/4/2015 2:44:33 PM		

- Refer the above power measurements done on 11 April 2015,

$$AN*A = 386.0235 \text{ VA}$$

$$BN*B = 836.114 \text{ VA}$$

$$CN*C = 939.174 \text{ VA}$$

$$939.174 + 836.114 + 386.0235 = 2161.3115 \text{ VA (Full machine apparent power)}$$

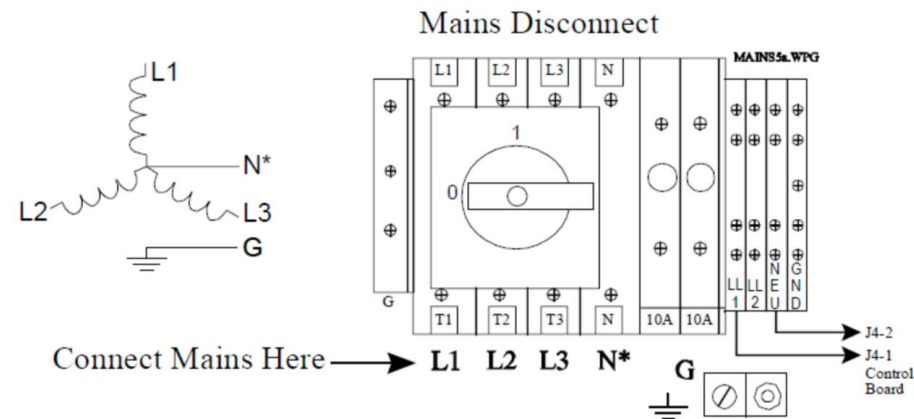
Using rule of thumb, taking V-line is for Wye with neutral 415V the I-line required is

$$2161.3116\text{VA} / 415\text{V} * 150\% = 7.82\text{A}$$

Taking the common next higher rating of fuse is 10A.

So it is stated that for Delta and Wye without Neutral the Max Amp Load = 17A. Refer the next for the relationship between 3-phase Wye with Neutral and (3-phase Delta and Wye without Neutral)

220/380–240/415 V 3-Phase Wye with Neutral



* If loads of 220-240 volts are required, the neutral must be connected.

Wiring the mains: 220/380—240/415V 3-phase wye with neutral

NOTE

The line-to-neutral voltage must not exceed 240 volts. The voltage between any two lines (line-to-line) may be as high as 415 volts.

Tools for connecting the mains and loads:

- 1/4-inch flat blade screwdriver for the Mains Disconnect Switch

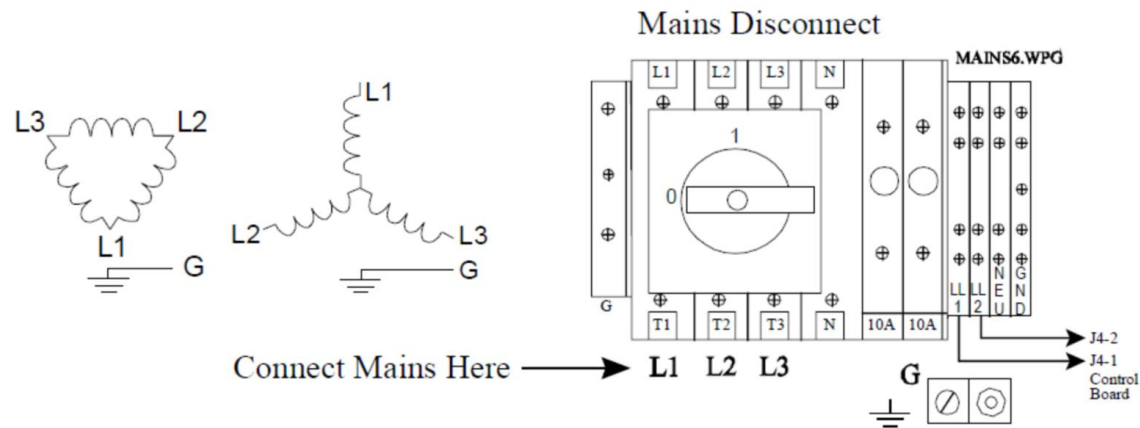
- 1/8-inch flat blade screwdriver for the Output Terminal Block

Set the Voltage Selector Switch (S1) on the Control Board to 230V

Wiring Loads: Line-to-Line (380-415V)
Line-to-Neutral (220-240V)

Wiring the AC Mains and Loads to the PDU

200–240 V 3-Phase Wye or 3-phase Delta



Wiring the mains: 200 – 240 V 3-phase wye or 3-phase delta

Tools for connecting the mains and loads:

- 1/4-inch flat blade screwdriver for the Mains Disconnect Switch
- 1/8-inch flat blade screwdriver for the Output Terminal Block

Set the Voltage Selector Switch (S1) on the Control Board to 230V

Wiring Loads: Line-to-Line (200-240 V)

- For Wye (with Neutral), the current across the lines is the same as the phase current:

$$I\text{-line} = I\text{-phase}$$

- Let's take that the machine max load runs at approximately 4000VA and $I\text{-phase} = 4000\text{VA}/415\text{V} = 9.63\text{A}$ (~10A).

- For Wye (without Neutral) and Delta source, the current across the lines is 1.732 times the phase current supplied to the loads:

$$I\text{-line} = 1.732 * I\text{-phase}$$

- Again let's take that the machine max load runs at approximately 4000VA and $I\text{-line} = 4000\text{VA}/240\text{V} = 16.67\text{A}$ (~17A). Which also meets the formula above where $1.732 * 9.63\text{A}$ (I-phase/line for Wye with Neutral) = 16.68A (I-line for Delta and Wye without Neutral)